FAIR data in the Wikibase Ecosystem A full-day workshop organized by DARIAH-EU WG DHwiki and Wikimedia Portugal

Assis, Tiago
dhwiki@dariah.eu
University of Porto

Barbosa, André
dhwiki@dariah.eu
Wikimedia Portugal

Candela, Gustavo
gcandela@dlsi.ua.es
University of Alicante

Hinzmann, Maria
hinzmannm@uni-trier.de
Trier Center for Digital Humanities, University of Trier

Joaquim, Manuel
mrocha@letras.up.pt
University of Porto

Kristen, Maximilian
Max.Kristen@kunstgeschichte.uni-muenchen.de
Ludwig-Maximilians-Universität München

Limão, Filomena
flimao@letras.up.pt
University of Porto

Lindemann, David
david.lindemann@ehu.eus
UPV/EHU University of the Basque Country

Malínek, Vojtěch
malinek@ucl.cas.cz
Institute of Czech Literature, Czech Academy of Sciences

Moitinho de Almeida, Vera
vmoitinho@letras.up.pt
University of Porto

Pellizzari di San Girolamo, Camillo Carlo
camillo.pellizzaridisangirolamo@sns.it
Scuola Normale Superiore Pisa

Salgado, Ana
anasalgado@fcsh.unl.pt
NOVA FCSH, Lisbon

Schöch, Christof
schoech@uni-trier.de
Trier Center for Digital Humanities, University of Trier

Silva, Carlos
carlos.silva@wikimedia.pt
Wikimedia Portugal

Trigo, Luis
ltrigo@letras.up.pt
University of Porto

Umerle, Tomasz
tomasz.umerle@ibl.waw.pl
Institute of Literary Resarch, Polish Academy of Sciences

Varvantakis, Christos
christos.varvantakis@wikimedia.de
Wikimedia Deutschland

Introduction

An increasing number of GLAM and DH community members are contributing to or using Wikidata for different purposes (Zhao 2022). In addition, Wikibase has recently emerged as a software solution chosen for a wide range of organizations and projects of different kinds and scales (Rossenova et al. 2023 , Shimizu et al. 2024). Wikibase use cases in the DH/GLAM context are manifold, and many projects rely on that software, some at a high level of maturity, some at an experimental stage. Self-hosting a Wikibase (Wikibase Suite) is not trivial, and demands system administrator staff, which may hinder smaller projects from using it. To address this, WMDE offers a free hosting solution for Wikibase (Wikibase

Cloud), which frees the user from the burden of system-admin tasks; the service allows anyone to have up to six Wikibase instances. This infrastructure makes deploying Wikibase and related tools even more interesting for GLAM institutions of different sizes and for researchers. For this reason, the wikibase.cloud service will be presented in the mini-conference as a low-barrier option for using the Wikibase infrastructure yourself.

Among the existing examples for GLAM institutions and DH research groups using Wikibase and Wikidata for different purposes are the following: MiMoText (University of Trier, a research project in computational literary studies focusing on the French Enlightenment novel); The Rematriation Project (supports Inuit Arctic communities in digitizing and sharing their cultural, tribal, and scientific knowledge); The Digital Archive of Artists' Publishing (interactive, user-driven, searchable database of artists' books and publications); e-Viterbo (supports academic research in the fields of art and architectural history); Digital Scriptorium (builds an online national catalog for premodern manuscripts in US collections) Europeana EAGLE (develops a multilingual online collection of millions of digitized items from museums, libraries, and archives which deal with inscriptions from the Greek and Roman World); FactGrid (a large collaborative database for historians); Rhizome Artbase (a Wikibase of artwork metadata); LexBib (a domain knowledge graph for lexicography and dictionary research built upon a digital bibliography); database of the Department of Digital Humanities of ELTE (Hungary: also deploys Wikibase). Recently, lexicographical (dictionary content) and text corpus data have also been modeled not only on Wikidata, but also on custom Wikibase instances (see Wikibase-Lex). Wikibase World contains a directory of existing and historical Wikibase instances, while in our group's own Wikibase Cloud instance, DHwiki , we document and curate existing research and scholarly publications about Wikibase and Wikidata. As for reports on conversions of legacy bibliographical catalogs to Wikibase LOD, we refer e.g. to recent experiments at OCLC (Bahnemann et al. 2021) and in the framework of the DARIAH WG Bibliodata (Lindemann et al. 2023).

The mini-conference has been scheduled for Monday, July 14, 9:00-17:00, with presentations targeting a broad public in the morning, and an afternoon session that will provide a deeper introduction to technical details and that primarily targets participants who are using or plan to use the Wikibase software themselves.

Wikidata / Wikibase tutorial

The tutorial section is designed for DH researchers, GLAM professionals, and Wikimedia contributors, whether experienced or new. Participants will develop an understanding of the functionalities of these tools, learn from real-world use cases, and explore practical workflows for managing and querying data. See the appendix for the planned structure of the session.

Wikibase use cases

In recent years, a wide range of institutions, organizations, groups and individuals from the Arts and Humanities have adopted Wikibase to host their databases. From datasets established for research purposes to databases aiming at the wider public, the Wikibase ecosystem embraces numerous and diverse use cases. Five different Wikibase use cases will be presented (see appendix).

DARIAH DHwiki Working group presentation, Q&A and Wikibase Onboarding session

In this third section, all presenters of the preceding two sessions will be available. The section will be introduced by presenting the new DARIAH working group DHwiki, and guidelines for getting started with the Wikibase Cloud service (see below). After that, participants will receive support for setting up their own Wikibase instance, and guidance for starting to upload their own datasets. The remainder will consist of all presenters answering the participants' questions, also regarding their presentations in the previous sessions, and giving hands-on support upon request. This will be supported remotely by the Wikibase Cloud team in Berlin.

Workshop Programme

Introduction

Structure and contents of the mini-conference, short presentation round of participants (*David Lindemann, Gustavo Candela)*

Wikibase tutorial

WikiFAIR: A guide for DH projects to better use Wikimedia technologies (*Maximilian Kristen)*

WikiFAIR is a proposal designed to help researchers make better use of existing Wikimedia infrastructure and tools. It emphasizes the collection of usage examples from ongoing projects and categorizes their methods into distinct, reusable operations. These operations are thoroughly described and documented within WikiFAIR, aiming to provide the scientific community with a clearer understanding of both the opportunities and challenges—technical and social—associated with leveraging Wikimedia tools and platforms in their work.

The Wikibase Ecosystem and Wikibase use cases overview (*Christos Varvantakis)*

In line with the Wikibase ecosystem joint vision , in recent years, a wide range of institutions, organizations, groups and individuals from the Arts and Humanities have adopted Wikibase to host their databases. From datasets established for research purposes, to databases aiming at the wider public, the Wikibase ecosystem embraces numerous and diverse use cases. In this introductory talk, we will discuss both the big picture of the use cases in the ecosystem as painted through various surveys and research, as well as we will have some more in-depth discussion of wikibase showcases.

How data is modeled, uploaded, edited, and queried on a Wikibase (*David Lindemann)*

Wikibase enables the collaborative creation of structured datasets in the shape of a Knowledge Graph; it combines interface-based editing and different methods of batch upload and edit while keeping a detailed and reversible edit history. Workflows and related tools known from Wikidata are mostly deployable for any Wikibase instance, which includes all entity types (“items” describing entities or concepts, properties, lexemes, and also entity schemes). We present how Wikibase stores and represents data, and how data is queried.

Federated queries in the Wikibase ecosystem (*Maximilian Kristen)*

Federated queries are a cornerstone of a functional linked open data future, yet they remain largely theoretical for many Wikibase projects. Considering federation as a feature early, before finalizing the ontology of your own Wikibase, is crucial to simplifying the process down the line. During our workshop in Munich in early December 2024, we updated the relevant documentation and will now offer an improved set of examples and guidelines to support successful inter-Wikibase operations.

Wikibase use cases

DataLib: Towards a prosopographical database about the network of Libanius (Camillo Carlo Pellizzari di San Girolamo)

DataLib (https://datalib.wikibase.cloud/) is a Wikibase instance about the 1544 Epistles of the rhetorician Libanius of Antioch (314–393), the biggest collection of epistles preserved from the Antiquity. It contains the numbers attributed to the Epistles in their editions and translations, and the dates (converted into EDTF format) attributed to them in the Foerster edition (Foerster 1921-1922), which is still the critical edition of reference today. The curators are now adding new statements to provide a coherent classification of the letters by topic and, most importantly, to connect each letter to all the people it mentions, in order to construct a reliable prosopographical database about the network of Libanius.

ENEOLI Wikibase: Connecting bibliography and meta-terminology of the domain of Neology (David Lindemann, Ana Salgado)

This Wikibase instance (https://eneoli.wikibase.cloud) serves as a collaborative platform for developing a multilingual vocabulary for neology. It combines conceptual tries with lexical entries (terms in multiple languages) identified in scientific articles from the domains of lexical innovation, lexicography and terminology. Articles are collected in Zotero, along with their publication metadata and full texts. These are processed to extract relevant terminological data, which is then added as content-describing metadata describing the publications in Wikibase. This setup enables advanced queries, such as analyzing term co-occurrence, conceptual evolution and term usage over time, and examining relationships between authors and specific terminology, while supporting ongoing collaboration on a living resource, involving all ENEOLI working languages.

Wikibase/Wikidata as a research tool in the classroom (Luís Trigo, Vera Moitinho de Almeida, Tiago Assis, Filomena Limão, Manuel Joaquim, André Barbosa, Carlos Silva)

Starting 2023, Wikibase has been deployed in different courses at the University of Porto; that initiative has resulted in several instances of open high-quality data, and a flourishing citizen science environment. Some examples are CreoPhonPt, Opustessellatum, Baroque Oporto, and BioImages.

Mapping DH Wikibase (Christof Schöch, Maria Hinzmann)

At the Trier Center for Digital Humanities, we're continuing the integration of LOD into our toolbox that we have started with MiMoText . One among several use cases using Wikibase is Mapping Digital Humanities , an initiative whose goal is to document the institutional history of Digital Humanities and to serve as a hub for information about institutions, events, publications and people associated with Digital Humanities. Development of this resource, both in terms of the data model and the modeled instances, is ongoing. A feature we will highlight because of its relevance to similar endeavors is our use of EntitySchemas to (partially) document our data model in a machine-readable manner and to ensure the consistency of our data.

Wikidata as the Central Provider of PIDs in the European Literary Bibliography (ELB) (Vojtěch Malínek, Tomasz Umerle)

The European Literary Bibliography (ELB) is a platform that collects, harmonizes, and presents bibliographic data on literature and literary studies from various national sources. ELB aims to integrate data from open resources provided by European providers. Currently, the platform includes data from the Czech Republic, Finland, Poland, and Spain. To enrich the data and ensure consistency, ELB uses Wikidata as a central source of persistent identifiers (PIDs). This approach links bibliographic records in ELB to the wealth of information available in Wikidata, particularly in the area of shared authority files. Integration with Wikidata not only enhances data consistency, but also facilitates statistical analyses and enables the development of analytical tools for exploring the entire ELB database.

DARIAH DHwiki Working group presentation, Q&A and Wikibase Onboarding session

DARIAH Working Group DHwiki, a new community space (*David Lindemann, Gustavo Candela, other group members)*

Since September 2024, this working group has been a common form of individuals interested in Wikidata and Wikibase coming from GLAM, DH research, and Wikimedia backgrounds. The group maintains a Wikibase instance and a Zotero group , and aims at being present and available for the community in different fora. We briefly introduce the group, the members, and their communication channels.

Presentation of the Wikibase Cloud service, from the user perspective (*Christos Varvantakis, David Lindemann*)

How do you set up an account for Wikibase Cloud? How do you create an instance? And most importantly, how do you establish the basics of modeling your database and start populating it? This presentation will address such questions, alongside a discussion of the origins, the basic functionalities and some use cases of Wikibase Cloud and related tools

Q&A and Wikibase Onboarding session (*DHwiki group members, supported remotely by the Wikibase Cloud team in Berlin*)